Day 080 | Unit 06 | Teacher plan

Understand and Use Ratios and Rates

Explain Equivalent Ratios in Two Models

Standards: 6.RP.A.3a

Learning target: I can explain Equivalent Ratios in Two Models, and explain and check my reasoning.

enVision Topic 5: Understand and Use Ratios and Rates | Spectrum companion Chapter 3

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Multiply both values in a ratio by the same scale factor. Organize pairs in a table and check that every row compares the same quantities in the same order. For a ratio a:b, draw a sections for one quantity and b sections for the other. A total covers a+b sections. Divide the total by a+b to find one section, then multiply by each part count.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 2

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01 / Learn

Multiply both values in a ratio by the same scale factor. Organize pairs in a table and check that every row compares the same quantities in the same order. For a ratio a:b, draw a sections for one quantity and b sections for the other. A total covers a+b sections. Divide the total by a+b to find one section, then multiply by each part count.

Worked example 1

A mix uses 6 cups of oats for 6 cups of seeds. Complete the table (oats, seeds): (6, 6), (12, ___), (18, ___).

1. Multiply both values in a ratio by the same scale factor.

2. Organize pairs in a table and check that every row compares the same quantities in the same order.

3. Multiply both parts by 2 and by 3.

4. State the result in the form requested: 12; 18.

Answer: 12; 18

Worked example 2

The ratio of paper flags to fabric flags is 3:3. There are 12 flags. Part A: draw a tape diagram and enter each count. Part B: a student says there are 7 paper flags. Use your model to explain the error.

1. For a ratio a:b, draw a sections for one quantity and b sections for the other.

2. There are 6 equal sections.

3. Each is 12 ÷ 6 = 2.

4. Multiply the section counts by 2.

Answer: 6 paper; 6 fabric; the claim is 1 paper flag too many.

Day 080 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

A mix uses 6 cups of oats for 3 cups of seeds. Complete the table (oats, seeds): (6, 3), (12, ___), (18, ___).

02 / Constructed response

The ratio of oak trees to pine trees is 7:6. There are 52 trees altogether. Draw a tape diagram and find each count.

03 / Constructed response

A mix uses 6 cups of oats for 6 cups of seeds. Complete the table (oats, seeds): (6, 6), (12, ___), (12, ___).

04 / Constructed response

The ratio of oak trees to pine trees is 8:3. There are 22 trees altogether. Draw a tape diagram and find each count.

05 / Constructed response

A mix uses 3 cups of oats for 2 cups of seeds. Complete the table (oats, seeds): (3, 2), (6, ___), (18, ___).

06 / Constructed response

The ratio of oak trees to pine trees is 4:7. There are 44 trees altogether. Draw a tape diagram and find each count.

07 / Constructed response

A mix uses 5 cups of oats for 2 cups of seeds. Complete the table (oats, seeds): (5, 2), (10, ___), (15, ___) and explain how you know the last two rows are equivalent.

08 / Model and error analysis

The ratio of paper flags to fabric flags is 5:2. There are 21 flags. Part A: draw a tape diagram and enter each count. Part B: a student says there are 16 paper flags. Use your model to explain the error.

09 / Multiple select

A recipe uses 6 cups oats for 4 cups seeds. Select all (oats, seeds) pairs that keep this ratio.
A. (25, 16)
B. (12, 8)
C. (12, 9)
D. (18, 12)

10 / Constructed response

The ratio of oak trees to pine trees is 3:6. There are 36 trees altogether. Draw a tape diagram and find each count and the difference between the counts.

Day 080 | Unit 06 | Answer key

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1. 6; 9

Multiply both parts by 2 and by 3.

6.RP.A.3a | Constructed response

2. 28 oaks and 24 pines

13 equal sections total; each section is 52 ÷ 13 = 4.

6.RP.A.3a | Constructed response

3. 12; 12

Multiply both parts by 2 and by 2.

6.RP.A.3a | Constructed response

4. 16 oaks and 6 pines

11 equal sections total; each section is 22 ÷ 11 = 2.

6.RP.A.3a | Constructed response

5. 4; 12

Multiply both parts by 2 and by 6.

6.RP.A.3a | Constructed response

6. 16 oaks and 28 pines

11 equal sections total; each section is 44 ÷ 11 = 4.

6.RP.A.3a | Constructed response

7. 4; 6

Multiply both parts by 2 and by 3.

6.RP.A.3a | Constructed response

8. 15 paper; 6 fabric; the claim is 1 paper flag too many.

There are 7 equal sections. Each is 21 ÷ 7 = 3. Multiply the section counts by 3.

6.RP.A.3a | Model and error analysis

9. B, D - (12, 8); (18, 12)

Equivalent ratios multiply both quantities by the same factor. A uses 2; B uses 3. C and D change only one scaled quantity.

6.RP.A.3a | Multiple select

10. 12 oaks and 24 pines; difference 12

9 equal sections total; each section is 36 ÷ 9 = 4.

6.RP.A.3a | Constructed response